

Floppy 80 (M1) User Guide

Welcome to the Floppy 80 (M1) User Guide. This document will walk you through configuring, operating, and debugging your Floppy 80 emulation system.

Configuration

The Floppy 80 (M1) requires a **FAT-formatted SD card** to store disk images and configuration files.

All configuration files must be placed in the **Root folder** of the SD card. Subfolders are not supported, with the single exception of the /FMT directory.

1. system.cfg

This optional file defines global hardware settings. If a setting is omitted, its default value is used.

- MEM: Enables or disables the onboard 32KB RAM expansion. (1 = Enabled [Default], 0 = Disabled)
- VHD: Enables or disables Virtual Hard Disk support. (1 = Enabled [Default], 0 = Disabled)
- WAIT: Enables or disables CPU wait states. (1 = Enabled, 0 = Disabled [Default])

Example system.cfg:

```
MEM=1  
VHD=0  
WAIT=0
```

A Note on Wait States: Wait states slow down the Z80 CPU to ensure it waits for the floppy disk controller emulation to complete. While generally unnecessary for standard systems, they can help with overclocked CPUs where the processor outpaces the emulation. Note that wait states are known to disrupt critical, tightly timed native operations like formatting a physical floppy disk.

2. boot.cfg

This file tells the Floppy 80 which configuration profile (.INI) to load automatically when the device boots or resets. It must contain **only** the exact filename of your chosen INI file.

Example boot.cfg:

```
LD531.INI
```

3. .INI Files

These files define which virtual disk images map to your floppy and hard drives upon boot.

- Drive0 to Drive2: Specifies the .dmk floppy image for drives :0, :1, and :2.
- HD0 / HD1: Specifies the .hdv image for the primary and secondary virtual hard drives.

- Doubler: Enables or disables double-density floppy support. (1 = Enabled, 0 = Disabled)
-

Example .INI file:

```
Drive0=LD531-0.dmk
Drive1=LD531-1.dmk
Drive2=LD531-2.dmk
HD0=hard0.hdv
Doubler=1
```

File Types & Folders

- **.DMK Files:** Virtual floppy disk images using a standardized format compatible with common TRS-80 emulators.
- **.HDV Files:** Virtual hard drive images using the same standard format as FreHD and trs80gp.
- **/FMT Folder:** A dedicated root-level directory used strictly to hold blank, unmountable floppy images. These act as templates for the FDC FMT utility.

Operation

Important Power Sequence: Always power on the Floppy 80 *before* turning on the main TRS-80 computer. Power is supplied via the micro-USB port on the Pi Pico microcontroller. A built-in LED on the Floppy 80 indicates disk read/write activity.

The FDC Utility

The FDC command-line program allows you to interact with the Floppy 80 directly from inside the TRS-80 operating system. Versions are available for CP/M (Lifeboat and FMG) as well as most major TRSDOS-based environments.

- FDC STA — Displays the current system status and lists the contents of the active boot INI file.
- FDC DIR [filter] — Lists files in the SD card's root folder. Adding an optional text filter narrows down the results.
- FDC INI [filename.ini] — Switches the system to a new INI profile. If you omit the filename, a selectable menu of available INI files will appear, and your choice will be written to boot.cfg.
- FDC DMK [filename.dmk] [0/1/2] — Mounts a specific floppy image to drive :0, :1, or :2. Running FDC DMK without arguments opens an interactive selection menu.
 - *Example:* FDC DMK LDOS-DATA.DMK 2 mounts that file to drive :2.
- FDC FOR — An emergency backup formatting utility. It lets you select a blank template from the /FMT folder to overwrite and clear a currently mounted drive image if native OS formatting fails.
- FDC IMP [filename.ext] :n — Imports a raw file from the root of the SD card directly into virtual floppy drive n.

Operating System Compatibility

The following matrix details how various TRS-80 Model I operating systems interact with the Floppy 80 emulation layer.

Operating System	Native Formatting	Backup Utility	Notes / Special Requirements
CP/M 1.4.1 (Lifeboat)	Works	Works	Stable
CP/M 1.5 (FMG)	Works	Works	Stable
Double DOS 4.2.4	Works	Works	Stable
DOS Plus 3.50	Works	Conditional	Requires a double-sided .dmk template to format. BACKUP requires Doubler=0.
LDOS 5.3.1	Conditional	Works	Currently supports single-density only. Requires a system reboot after formatting a drive.
MultIDOS 4.01	Works	Conditional	BACKUP requires Doubler=0.
NEWDOS 3.0 / 80 V2.0	Works	Works	Stable
TRSDOS 2.3	Works	Works	Stable
UltraDOS 4.2.0	Works	Works	Stable
VTDOS 3.0.0	Works	Issue	BACKUP encounters "Different Pack IDs" warning.
VTDOS 4.0.1	Works	Conditional	BACKUP requires Doubler=0.

Debugging & Serial Interface

The Floppy 80 provides a Command Line Interface (CLI) over its USB connection for advanced diagnostics.

To access it, connect the Pi Pico to a host computer via USB and launch any standard serial communication application (such as PuTTY, Tera Term, or Serial). On macOS, the device usually initializes as a port named like `/dev/cu.usbmodem000000`. Because this is a virtualized serial bridge, baud rate and parity settings do not matter—just open the port and press **Enter** to see the help menu.

CLI Command Summary

Serial Command	Matching FDC Command	Description
<code>boot [file.ini]</code>	FDC INI	Sets the default INI configuration for boot or reset.
<code>disks</code>	<i>None</i>	Displays detailed geometry and mount status for all virtual drives.
<code>dir [filter]</code>	FDC DIR	Displays the SD card directory layout.
<code>dump [n]</code>	<i>None</i>	Performs a complete, raw sector-by-sector hexadecimal dump of drive n.
<code>hdc</code>	<i>None</i>	Formats/creates a virtual hard disk structure.
<code>help</code>	<i>None</i>	Displays the built-time CLI help menu.
<code>logon</code>	<i>None</i>	Enables real-time FDC transaction logging to the terminal screen.
<code>logoff</code>	<i>None</i>	Mutes real-time FDC transaction logging.
<code>status</code>	FDC STA	Returns the current global state of the device.

Advanced: Setting Up a Virtual Hard Disk (Beta)

Setting up a virtual hard disk (.hdv) requires configuring software drivers on your boot image and establishing rigid partition sizes.

Prerequisites

1. A working, bootable Drive 0 floppy image containing the standard **RSHARD** drivers (e.g.,

RSHARD5 for LDOS). These drivers can be found in the download section of Tim Mann's TRS-80 archive.

2. A blank .hdv hard disk image. The most reliable way to generate a fresh image container is using an emulator like **trs80gp**.

Hard Disk Partitioning Workflow (LDOS531 example)

Because traditional standard configurations (like 309 tracks, 4 heads) can cause size allocation bugs in emulated environments, you should use an intentional 4-partition geometry split of **38 cylinders per partition**. If you try to push a partition to 39 cylinders, the LDOS formatter will drop the drive down to a 35-track default layout.

Follow these configuration steps carefully inside your emulator before copying the file over to your physical Floppy 80 hardware:

1. **Initialize the Driver for Partition 1 (Drive :4)**

Boot your LDOS 5.3.1 image in the emulator. Execute the driver initialization command and provide the geometry values below when prompted:

```
SYSTEM (DRIVE=4,DISABLE,DRIVER="RSHARD5")
Enter drive address <1-2> and F/R: 1F
Enter the STEP RATE: [Press Enter for default]
Enter the physical TRACKS per surface: 153
Enter the total number of HEADS: 8
Enter partition's number of heads: 8
Enter partition's number of cylinders: 38
```

2. **Initialize Partition 2 (Drive :5)**

Bind the second virtual drive slice using the remaining available head allocation:

```
SYSTEM (DRIVE=5,DISABLE,DRIVER="RSHARD5")
Enter partition's number of heads: 8
Enter partition's number of cylinders: 38
```

3. **Initialize Partition 3 (Drive :6)**

Bind the third virtual drive slice:

```
SYSTEM (DRIVE=6,DISABLE,DRIVER="RSHARD5")
Enter drive address <1-2> and F/R: 1F
Enter partition's number of heads: 8
Enter partition's number of cylinders: 38
```

4. **Initialize Partition 4 (Drive :7)**

Bind the final virtual partition slice:

```
SYSTEM (DRIVE=7,DISABLE,DRIVER="RSHARD5")
Enter drive address <1-2> and F/R: 1F
Enter partition's number of heads: 8
Enter partition's number of cylinders: 38
```

5. **Format the New Partitions**

Run the Western Digital high-level formatter utility on each allocated drive. Repeat this process for drives :5, :6, and :7.

```
RSFORM5 :4
```

```
Disk Pack name? HD04
```

```
Master password? PASSWORD
```

```
Are you sure you want to format it <Y,N>? Y
```

```
Lock out track manually <Y/N>? N
```

6. **Commit Configuration via Sysgen**

Save your newly assigned drive mappings so they survive system restarts by running:

```
SYSTEM (SYSGEN)
```

Deploying to the Floppy 80

Once configured via the emulator, move your files onto the physical SD card and ensure your configuration parameters match the layout below:

Create an INI profile (e.g., ldosboothd.ini):

```
Drive0=ldosboot2s.dmk
```

```
HD0=ldosboothd.hdv
```

Ensure that VHD=1 is specified within your global system.cfg, and check that your boot.cfg points directly to your newly created hard disk .ini profile. Once booted, typing FREE or DEVICES at the LDOS prompt will verify your active virtual hard disk sub-systems.